

1. What type of TV screen technologies are currently available in Australia and which are the most frequent?

What data is needed?

- **Screen_Tech:** Categorical data (String).
- **Count:** Discrete numerical data (Integer).

What chart type would be good and reasons? A **Bar Chart** (Vertical). While a pie chart was shown in your worked example, bar charts are objectively superior for comparing frequencies. The human eye is much better at comparing lengths (bars) than angles or areas (pie slices).

Data Table Sketch

Screen_Tech	Count
LCD (LED)	3367
LCD	433
OLED	284

Chart Sketch (Text Representation)

Plaintext

Count

3500 | [====]

3000 | [====]

2500 | [====]

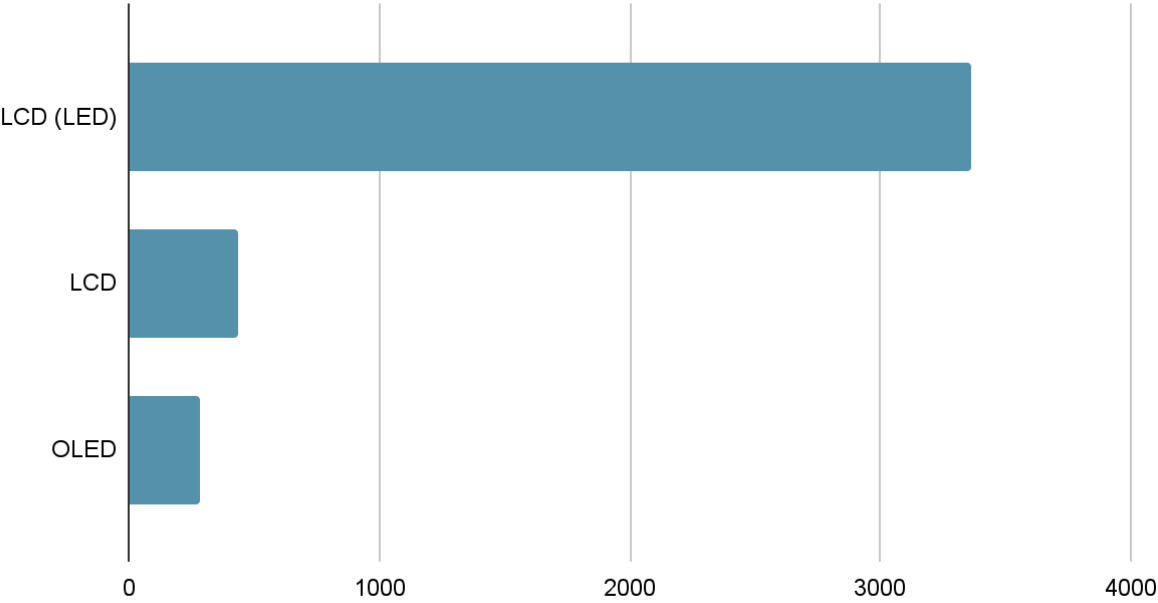
| ...

500 | [====] [====]

0 | [====] [====] [====]

LCD(LED) LCD OLED

Points scored



2. What screen sizes are currently available, and which are the most frequent?

What data is needed?

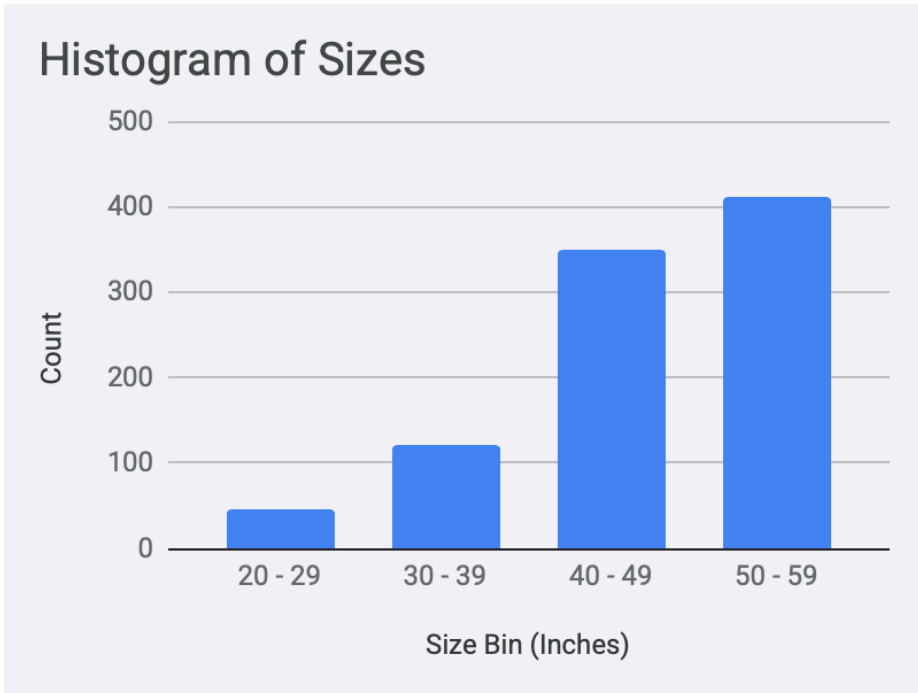
- **Screen_Size:** Continuous numerical data (e.g., inches or cm).
- **Frequency/Count:** Discrete numerical data (Integer).

What chart type would be good and reasons? A **Histogram**. Because screen size is a continuous variable, a standard bar chart is incorrect. A histogram bins the continuous data into ranges (e.g., 30-39", 40-49") to accurately show the distribution and central tendency of the sizes available.

Data Table Sketch (Binned Data)

Size_Bin (Inches)	Count
20 - 29	45
30 - 39	120
40 - 49	350
50 - 59	410

Chart Sketch



3. Which brands have the greatest number of different models?

What data is needed?

- **Brand:** Categorical data (String).
- **Model_Count:** Discrete numerical data (Integer).

What chart type would be good and reasons? A **Horizontal Bar Chart** sorted in descending order. When dealing with many categories (dozens of TV brands), a vertical bar chart causes text labels on the X-axis to overlap and become unreadable. A horizontal layout ensures all brand names are legible.

Data Table Sketch

Brand	Model_Count
Samsung	145
LG	132

Hisense	89
Sony	76

Chart Sketch (Text Representation)

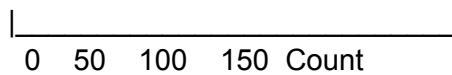
Plaintext

Samsung | [=====] 145

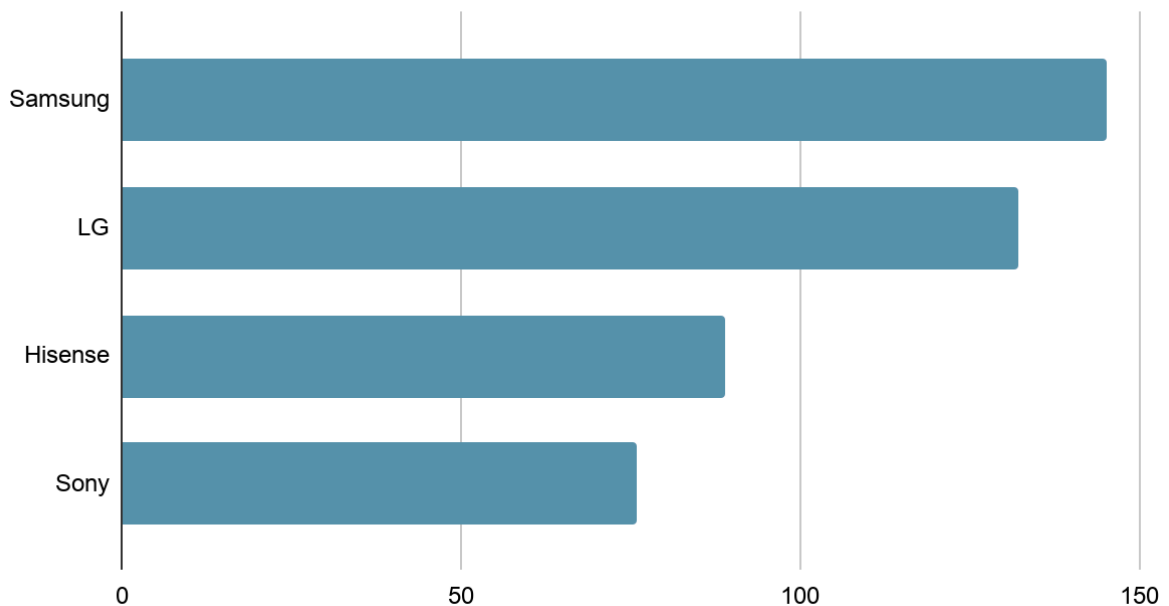
LG | [=====] 132

Hisense | [=====] 89

Sony | [=====] 76



Horizontal Bar Chart



4. Which type of screen technology consumes the least amount of power?

What data is needed?

- **Screen_Tech:** Categorical data (String).

- **Power_Consumption:** Continuous numerical data (Float/Integer).

What chart type would be good and reasons? A **Box Plot**. Relying on simple averages (means) is highly susceptible to outliers (e.g., a few massive 85" TVs will drastically skew the average power consumption of LED tech). A box plot accurately displays the median, the interquartile range (where most data falls), and flags outliers, giving a rigorous view of actual efficiency.

Note: To build a box plot in KNIME, you do not aggregate the data first. You feed the raw data table directly into the Box Plot node.

Data Table Sketch (Raw Format required for Box Plot)

Row ID	Screen_Tech	Power_Consumption (kWh)
Row0	OLED	320
Row1	LCD	410
Row2	OLED	315

Chart Sketch

Row ID	Screen_Tech	Power_Consumption (kWh)
Row0	OLED	320
Row1	LCD	410
Row2	OLED	315

5. What is the relationship between screen size and power use?

What data is needed?

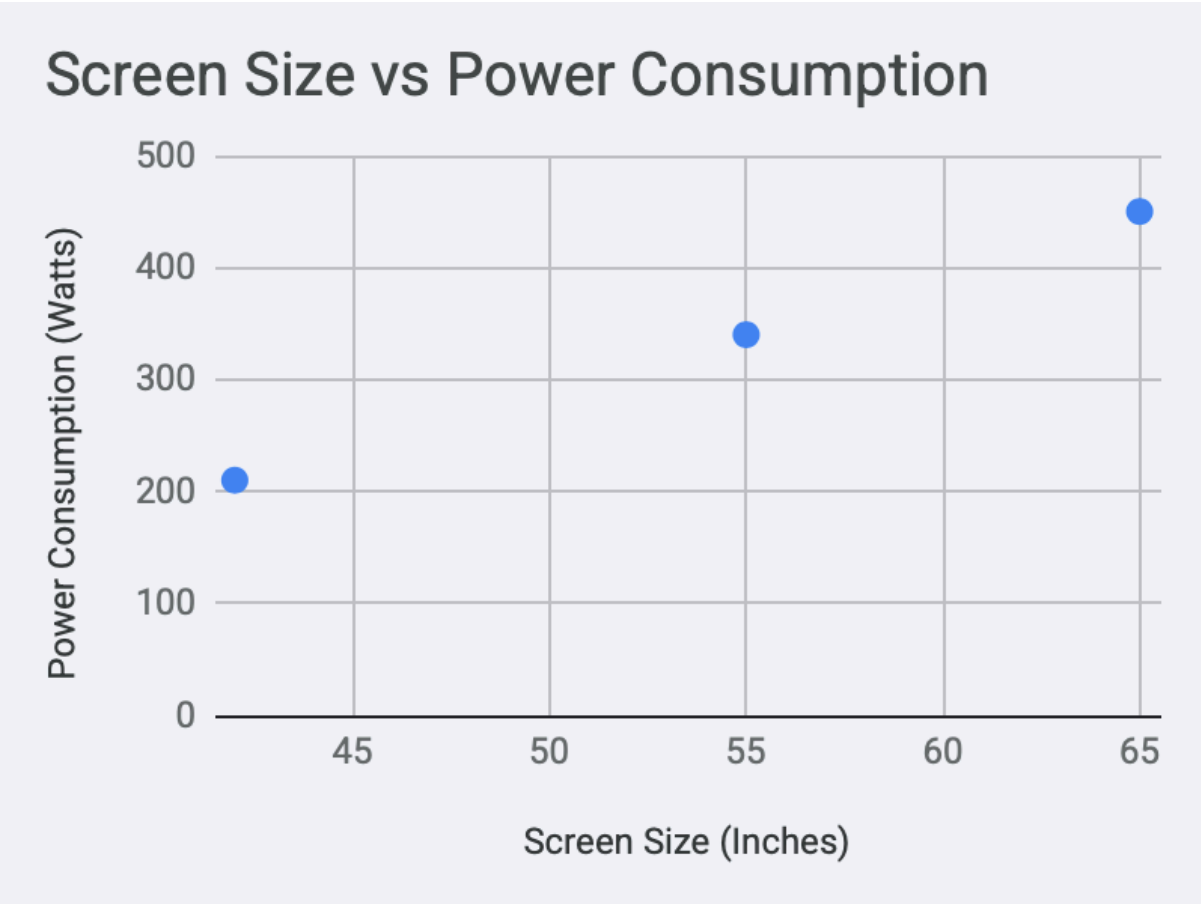
- **Screen_Size:** Continuous numerical data.
- **Power_Consumption:** Continuous numerical data.

What chart type would be good and reasons? A **Scatter Plot**. This is the definitive chart type for analyzing the correlation or relationship between two continuous numerical variables. It will instantly reveal if larger screens strictly equate to higher power use, and if that growth is linear or exponential.

Data Table Sketch (Raw Format)

Row ID	Screen_Size	Power_Consumption
Row0	42	210
Row1	55	340
Row2	65	450

Chart Sketch (Text Representation)



6. What is the relationship between star rating and screen size?

What data is needed?

- **Star_Rating:** Ordinal categorical data (e.g., 1, 1.5, 2... 6).
- **Screen_Size:** Continuous numerical data.

What chart type would be good and reasons? A **Grouped Box Plot** (grouped by Star Rating). Because star ratings are discrete, tiered categories, a scatter plot will result in heavy overlapping points at specific star intervals. A box plot for each star rating category will clearly show how the distribution of screen sizes changes as the efficiency rating increases or decreases.

Data Table Sketch (Raw Format)

Row ID	Star_Rating	Screen_Size
Row0	3.5	55

Row1	4.0	42
Row2	2.5	65

Chart Sketch

